Week 19: No card inside another

Adult guide built from the problem files · Unpiloted; card making and the launch unrehearsed.

This week's adult guide is week-19-facilitator.pdf, whose source is editable/facilitator-src/guide.json. These problem files hold each problem's one-sentence idea; the full keys, hints and certificates are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Four small groups	P1			first
Largest collections from the four- and eight-card decks			P1	first
Every collection from four cards	P2	P1		first
Search the sixteen-card deck (Grades 4-5)			P2	first
Two-card collections from eight cards	P3			first
The largest collections from eight cards	P4	P2		first
Keep the starting cards (eight-card deck)	P5	P3		K-1 return; 2-3 first
Keep the starting cards (sixteen-card deck)			P3	first
Fewest nested rows for the four- and eight-card decks	P6	P4	P4	return
Could four cards work?	P7	P5		return
Set aside any one card	P8			return
Search the sixteen-card deck (Grades 2-3)		P6		return
Fewest nested rows for sixteen cards		P7	P5	return
Prove the largest size from sixteen cards		P8	P6	return
Largest collection with the circle card			P7	return
Every largest collection			P8	return

First route. K–1 P1–P4; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P5–P8; Grades 2–3 P4–P8; Grades 4–5 P4–P8.

Key, one entry per problem

Four small groups

K–1 P1 · first route

Idea. Overlapping cards can share a collection; a card holding another cannot, and the empty card fits inside every card.

Largest collections from the four- and eight-card decks

Grades 4–5 P1 · first route

Idea. The maxima are 2 and 3, and a cover by that many nested rows proves no larger collection works.

Every collection from four cards

K–1 P2 · Grades 2–3 P1 · first route · wording differs by band

Idea. There are exactly five: each card alone and A/B; empty and AB can join no other card.

Search the sixteen-card deck (Grades 4-5)

Grades 4–5 P2 · first route

Idea. The six two-symbol cards form a collection; whether seven is impossible is the question to settle.

Two-card collections from eight cards

K–1 P3 · first route

Idea. Nine pairs work: two singletons, two pairs, or a singleton with the pair that lacks its symbol.

The largest collections from eight cards

K–1 P4 · Grades 2–3 P2 · first route · wording differs by band

Idea. The maximum is three, reached only by A/B/C and AB/AC/BC.

Keep the starting cards (eight-card deck)

K–1 P5 · Grades 2–3 P3 · K-1 return; 2-3 first route · wording differs by band

Idea. ABC and A/BC take no more cards, yet three is possible: being unable to add a card is not the same as largest.

Keep the starting cards (sixteen-card deck)

Grades 4–5 P3 · first route

Idea. ABCD and A/BCD are unextendable at sizes 1 and 2, while AB/AC extends to six.

Fewest nested rows for the four- and eight-card decks

K–1 P6 · Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. Two and three rows suffice and are needed, since the one-symbol cards must go in different rows.

Could four cards work?

K–1 P7 · Grades 2–3 P5 · return route · wording differs by band

Idea. No: three nested rows cover the deck, so any four cards put two in one row.

Set aside any one card

K–1 P8 · return route

Idea. Yes: A/B/C and AB/AC/BC share no card, so removing one card leaves one of them whole.

Search the sixteen-card deck (Grades 2-3)

Grades 2–3 P6 · return route

Idea. The six two-symbol cards form a collection of six; the rows that follow show it is best.

Fewest nested rows for sixteen cards

Grades 2–3 P7 · Grades 4–5 P5 · return route · wording differs by band

Idea. Six nested rows cover the sixteen cards, and the six two-symbol cards need six different rows.

Prove the largest size from sixteen cards

Grades 2–3 P8 · Grades 4–5 P6 · return route · wording differs by band

Idea. Six is the largest: the two-symbol cards reach it and a six-row cover allows at most one card per row.

Largest collection with the circle card

Grades 4–5 P7 · return route

Idea. Four: every other card must avoid the circle, leaving a three-symbol deck whose maximum is three.

Every largest collection

Grades 4–5 P8 · return route

Idea. The six two-symbol cards are the only six-card collection: singletons and, by complements, triples limit a collection to four.